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Studierichting: Informatica

Oefeningenreeks 4A nr. 44.4

$$\int \frac{x^4 + x^3 - 3x + 5}{x^3 + 2x^2 + 2x + 1} dx$$

$$x^3 + 2x^2 + 2x + 1 = (x + 1)(x^2 + x + 1)$$

$$\frac{x^4 + x^3 - 3x + 5}{x^3 + 2x^2 + 2x + 1} = x - 1 + \frac{6 - 2x}{x^3 + 2x^2 + 2x + 1}$$

$$\frac{6 - 2x}{(x + 1)(x^2 + x + 1)} = \frac{A}{x + 1} + \frac{Bx + C}{x^2 + x + 1}$$

$$A = \left. \frac{6 - 2x}{x^2 + x + 1} \right|_{x=-1} = 8$$

$$Bx + C = \left. \frac{6 - 2x}{x + 1} \right|_{x=-\frac{1}{2} + i\frac{\sqrt{3}}{2}}$$

$$= \frac{7 - i\sqrt{3}}{\frac{1}{2} + i\frac{\sqrt{3}}{2}}$$

$$= \frac{2(7 - i\sqrt{3})}{1 + i\sqrt{3}}$$

$$= \frac{2(7 - i\sqrt{3})(1 - i\sqrt{3})}{(1 + i\sqrt{3})(1 - i\sqrt{3})}$$

$$= \frac{2(4 - 8i\sqrt{3})}{4}$$

$$= 2 - 4i\sqrt{3}$$

$$\Rightarrow -\frac{B}{2} + C + i\frac{B\sqrt{3}}{2} = 2 - 4i\sqrt{3}$$

$$\Rightarrow \frac{B\sqrt{3}}{2} = -4\sqrt{3} \Rightarrow B = -8$$

$$\Rightarrow -\frac{8}{2} + C = 2 \Rightarrow C = -2$$

$$\Rightarrow \int \left(x - 1 + \frac{8}{x + 1} + \frac{-8x - 2}{x^2 + x + 1} \right) dx$$

$$= \int (x - 1) dx + 8 \int \frac{dx}{x + 1} - \int \frac{8x + 2}{x^2 + x + 1} dx$$

$$8x + 2 = 4(2x + 1) - 2$$

$$= \frac{x^2}{2} - x + 8 \ln|x + 1| - 4 \int \frac{2x + 1}{x^2 + x + 1} dx + 2 \int \frac{dx}{x^2 + x + 1}$$

$$= \frac{x^2}{2} - x + 8 \ln|x + 1| - 4 \ln(x^2 + x + 1) + 2 \int \frac{dx}{\left(x + \frac{1}{2}\right)^2 + \frac{3}{4}}$$

$$= \frac{x^2}{2} - x + 8 \ln |x + 1| - 4 \ln (x^2 + x + 1) + \frac{4}{\sqrt{3}} \operatorname{Bgtan} \left(\frac{2x + 1}{\sqrt{3}} \right) + k$$

References